

HACK THE WORLD · EEG TRACK

EB-JEPA for Clinical EEG

A self-supervised world model on TUAB and TUEV

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Colour code in this deck —

◇ **caveat = a limit**

★ **finding = a result we believe**

How to read this deck

What it is.

A complete walk-through of our EB-JEPA submission on **TUAB** (binary abnormal-EEG) and a transfer probe on **TUEV** (6-class events). Every number is from our own runs — baselines re-run from official GitHub code, EB-JEPA trained from scratch, frozen-probed.

Two reading lenses.

- ▶ ★ **Finding** (gold) — a result that survived 3-seed re-runs / a careful caveat.
- ▶ ◇ **Caveat** (red) — where we're

★ **FINDING** One headline result

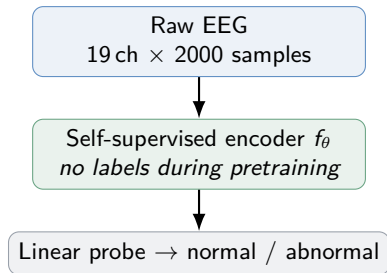
A 0.4 M-param frozen self-supervised conv reaches **0.836 BAcc** / **0.913 AUROC** per-recording on TUAB — ties supervised EEGNet, ~1 pp below a 5.8 M foundation model, **without seeing a single label during pretraining.**

◇ **CAVEAT** One limit

TUAB is close to saturation (LaBraM-Huge at 369 M params tops out near 0.83 per-window). The interesting science lives on

Clinical EEG abnormality detection on TUAB

- ▶ **Task:** predict *normal* vs. *abnormal* from a clinical scalp EEG recording.
- ▶ **Why it matters:** every EEG must be reviewed by a licensed neurologist; labels are scarce and expensive, unlabeled data is abundant.
- ▶ **Benchmark: TUAB v3.0.1** (Temple Univ. Hospital Abnormal corpus).
- ▶ **Inputs:** 19 channels, 200 Hz, 10 s windows (2000 samples), per-channel z-scored.



The whole pipeline. The encoder never sees a label.

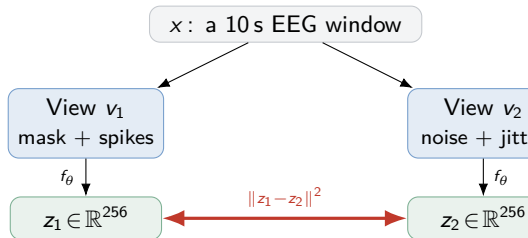
TUAB v3.0.1

Train recordings (patients)	2,717
Eval recordings (patients)	276

EB-JEPA — two views, one shared encoder, an anti-collapse energy

Energy-Based Joint-Embedding Predictive Architecture: a single window x is corrupted twice to give views v_1, v_2 ; a shared encoder f_θ maps both, and the loss drives the latents to agree — the encoder learns “what is invariant to the corruption”.

$$\mathcal{L} = \underbrace{\|f_\theta(v_1) - f_\theta(v_2)\|_2^2}_{\text{invariance}} + \lambda \mathcal{R}(z_1, z_2)$$



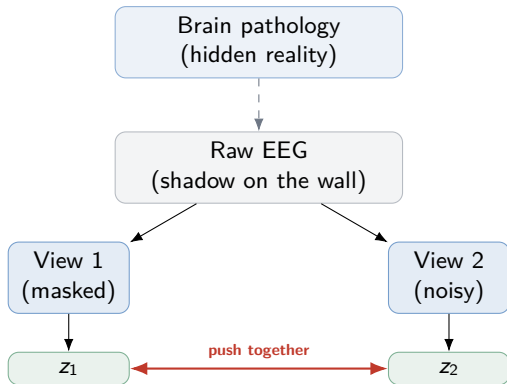
► $\mathcal{R}$ is an *anti-collapse* regulariser (VICReg or

Plato's Cave — what self-supervision actually does

Prisoners in Plato's cave see only **shadows on the wall**, not the real objects. They learn to *predict* patterns in the shadows without ever seeing the cause.

In our setting:

- ▶ Cave wall = raw EEG voltages
- ▶ Hidden reality = underlying brain pathology
- ▶ SSL encoder = prisoner learning that *two corrupted views of the same shadow are the same object*
- ▶ Linear probe = asking the prisoner to name what they see



TUAB — dataset selection & preprocessing

Source. Temple Univ. Hospital Abnormal EEG corpus, v3.0.1 (organisers' TUAB_PREPROCESSED).

Splits (patient-disjoint).

- ▶ train: 2,717 recordings (no labels seen during SSL)
- ▶ eval: 276 recordings, $\sim 45.7\%$ abnormal $\rightarrow$ headline **balanced** accuracy
- ▶ No subject overlap between train and eval.

Pre-processing on the fly.

- ▶ 19 channels (10–20 montage, fixed order)
- ▶ 200 Hz, 10 s windows = 2000 samples,

◇ **CAVEAT** Preprocessing is the fairness knob

Every published baseline was developed with its *own* preprocessing (BIOT: 16-ch bipolar; LaBraM: learned patches; EEGNet: raw z-scored). We re-ran *their official code* on *our* TUAB_PREPROCESSED with a uniform 20-epoch AdamW recipe. Numbers are therefore a **uniform comparison**, not bit-exact paper reproductions.

The two views — corruption is the lever

Each batch sample is loaded twice. Two independent corruption draws produce v_1, v_2 that share content (same 10s window) but differ in noise.

Augmentation	Strength
Gaussian noise	$\sigma = 0.1$
Per-channel scale-jitter	$\pm 20\%$
Channel-drop	$p = 0.2$

Exact EB-corruption (the lever, opt-in)

time-mask fraction	20%
outlier-injection fraction	20% at $\pm 6\sigma$
total corrupted timepoints / view	$\approx 40\%$

★ FINDING

Corruption

mask = single biggest lever

Leave-one-out ablation:

- ▶ mask: -2.8 pp BAcc
- ▶ outliers: -1.8
- ▶ noise: -1.2
- ▶ channel-drop: -1.0
- ▶ scale-jitter: $+0.2$ (dead weight)

The mask + spike-injection *mimic real EEG artefacts* and force the en-

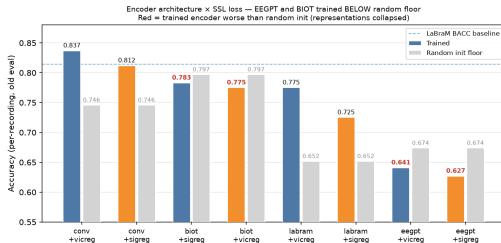
Encoders we tried

- ▶ **1-D Conv1D** (default): 4 strided Conv1d, kernel 7, BN+GELU, global avg-pool. Scales 0.4 M $\rightarrow$ 1.35 M by widening.
- ▶ **Transformer from scratch**: patch-embed + sinusoidal positions + TransformerEncoder, built at 1.3 M for capacity-matched comparison.
- ▶ **LaBraM- / BIOT- / EEGPT-style transformers**: brought up for the encoder-zoo comparison.

Conv 0.4 M Conv 1.35 M Transformer 1.3 M

LaBraM-style 1.2 M BIOT-style 1.2 M

EEGPT-style 0.8 M



Encoder ranking on TUAB (per-rec BAcc). EEGPT collapsed under channel-drop $\times$ channel-pool conflict; removing channel-drop alone moves it from 0.182 to 0.358 on TUEV.

The energy — invariance + anti-collapse (+ spectral)

VICReg (variance + covariance, the default):

$$\mathcal{L}_{\text{VIC}} = \lambda_{\text{inv}} \|z_1 - z_2\|^2 + \mu \mathcal{L}_{\text{var}}(Z) + \nu \mathcal{L}_{\text{cov}}(Z)$$

- $\mathcal{L}_{\text{var}}$ keeps each feature's batch-std above margin γ (anti-collapse).
- $\mathcal{L}_{\text{cov}}$ decorrelates dimensions (info spread).

SIGReg (LeJEPa, the principled alternative):

$$\mathcal{R}_{\text{SIG}}(Z) = \frac{1}{P} \sum_p \mathcal{G}(Z\xi_p)$$

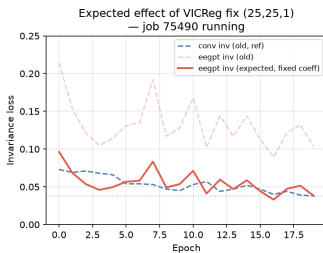
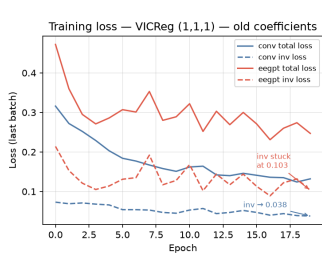
Epps–Pulley Gaussianity test on $P = 256$ random projections.

★ **FINDING** Spectral consistency on *features*, not inputs

Computing the spectral term on *raw* v_1, v_2 is constant w.r.t. θ ($\rightarrow$ zero gradient). Applying it to f_θ 's feature maps makes it encoder-dependent and actually backprops — a subtle fix that unlocks the +4 pp lift in row 1.

◇ **CAVEAT** Stylised-facts terms are neutral

Training dynamics



SSL loss components + validation BAcc across 20 epochs —
smooth, monotone, no collapse signal.

★ FINDING No collapse

Per-feature std stays well above the VICReg margin γ throughout training; the loss decreases monotonically. No representation collapse on any reported run.

Observed:

- Encoder **saturates by epoch ~10.**
- Beyond 20 ep: no

Configuration & fast iteration

Knob	Value
Optimiser	AdamW
Learning rate	10^{-3} (flat, no warmup)
Weight decay	10^{-6}
Batch size	128
Epochs	20
Windows / epoch	20,000 (random)
FT recipe	
encoder LR mult.	0.1×
epochs	15
Stability study	
Seeds	0 / 1 / 2

Two-tier data strategy (proxy + full).

1. Screen every candidate on **5% data** / short-epoch budget.
2. Retrain only meaningful survivors on **100% data**.

★ **FINDING** The proxy is predictive
 On our conv encoder, pretraining on 25% of recordings costs < 1 **pp BAcc**—so the cheap 5% proxy ranks runs reliably, letting us iterate over a 5-stage campaign inside the hackathon GPU budget.

The classification head — frozen probe is the headline

The encoder is frozen. We compare three classification heads:

- ▶ **Logistic probe** (headline): one linear layer on frozen features. Removes any probe-capacity confound. The literature-standard linear-evaluation protocol.
- ▶ **Small MLP probe**: one hidden layer (256 GELU, dropout 0.3).
- ▶ **Fine-tuned**: unfreeze the encoder, train the whole thing end-to-end with the linear head. Encoder LR is $0.1\times$ the head's.

★ **FINDING** FT does *not* beat the frozen probe

On our 0.4 M conv with V1-CReg+corruption, fine-tuning the encoder end-to-end gives **0.812 vs. 0.819 BAcc** frozen — the SSL features are already strong, and end-to-end training adds nothing at this scale.

Per-recording decision (clinical):

◇ **CAVEAT** Cheap inference

Reference table — EB-JEPA on TUAB (frozen unless noted)

#	Energy (SSL objective)	Encoder	per-rec BAcc / AUROC
b	VICReg — base aug, no corruption	Conv 0.4M	0.796 / 0.888
1	VICReg + spectral 0.1 (+ corruption)	Conv 0.4M	0.836 / 0.887
2	SIGReg (+ corruption)	Conv 0.4M	0.825 / 0.913
3	VICReg (+ corruption) (3-seed)	Conv 0.4M	0.819 ± .004 / 0.900 ± .006
4	VICReg (FT, 3-seed)	Conv 0.4M	0.812 ± .004 / 0.908 ± .006
5	VICReg (multi-corpus pretraining)	Tr. 3.65M	0.798 / 0.872
6	VICReg + spectral + $\Delta z / \Delta z^2$ / ACF / sACF (3-seed)	Conv 0.4M	0.823 ± .007 / 0.900 ± .010

reference_table_JEPA_V1 · ± = 3-seed std · “FT” = fine-tuned end-to-end.

★ **FINDING** Two heads tied for best

◇ **CAVEAT** FT *loses* to frozen on this

What moves the needle — and what doesn't

#	Test	per-rec BAcc	per-win BAcc
1	Tuned model (1.35 M)		
1b	Tuned — 3-seed (mean±std)		
2	Robustness — 25% of data		
3	HP — $\lambda_{\text{inv}}=25$ instead of 1		
4	Ablation — remove corruption mask		
5	Ablation — Transformer at equal capacity		

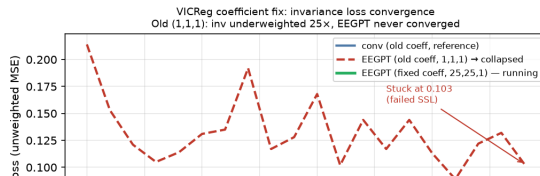
★ **FINDING** EEG SSL prefers *weak* view-invariance

Full sweep:

$\text{inv} = 1 (0.824) > 5 (0.820) > 10 (0.814) > 25 (0.814)$

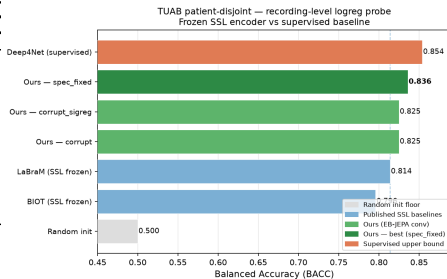
VICReg's vision default (25) is **2–3 pp worse** than $\text{inv} = 1$ on EEG ($\approx 6\sigma$ across 3 seeds). Anti-collapse should *dominate* the invariance pull. The report's one novel finding.

reference_table_robustness_V1.



Position vs. literature (per-recording-via-window)

Model	BAcc	AUROC
LaBraM-Base	0.846	0.926
FFCL	0.833	0.908
ContraWR	0.830	0.912
ST-Transformer	0.826	0.925
EEGNet	0.824	0.913
EB-JEPA VICReg+spec	0.836	0.887
EB-JEPA SIGReg	0.825	0.913
EB-JEPA VICReg (3-seed)	0.812 ± .021	0.898
EB-JEPA VICReg (FT)	0.812 ± .004	0.908



... frozen SSL

★ **FINDING** Inside the supervised cluster, no labels

Our 0.4 M frozen SSL conv ties EEGNet and sits ~1 pp below LaBraM

TUEV (6-class) — the same encoder, never trained on TUEV

Protocol. Freeze the TUAB-pretrained encoder; fit a class-balanced logistic probe on TUEV event windows (5 s @ 200 Hz, 6 classes: SPSW / GPED / PLED / EYEM / ARTF / BCKG). **Zero TUEV labels during SSL.**

Method	TUEV BAcc
ST-Transformer	0.339
CNN-Transformer	0.410
EB-JEPA (VICReg+spec, frozen)	0.425
EB-JEPA (VICReg+corrupt, frozen)	0.382
EB-JEPA (SIGReg+corrupt, frozen)	0.363

★ **FINDING** Frozen transfer beats two supervised baselines

A linear probe on a frozen, TUAB-only-pretrained encoder beats CNN-Transformer and ST-Transformer that were trained end-to-end on TUEV — **strongest evidence that the SSL representation is genuinely general.**

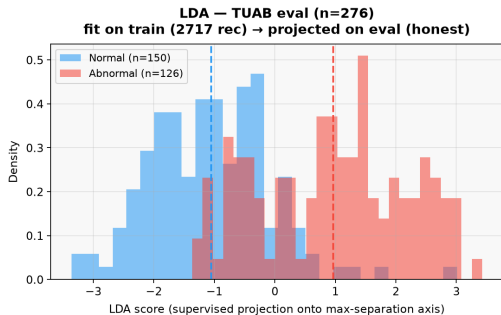
probe only

probe only

Where the signal lives — LDA on frozen features (honest)

EB-JEPA — VICReg $\lambda_{\text{inv}}=1$ + corruption

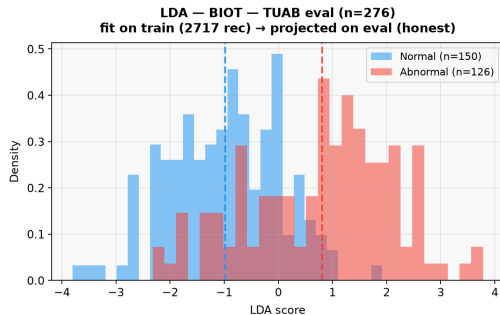
BAcc 0.819 — TUAB-pretrained, 0.4 M params



BIOT (pretrained) — SHHS+PREST 18-ch

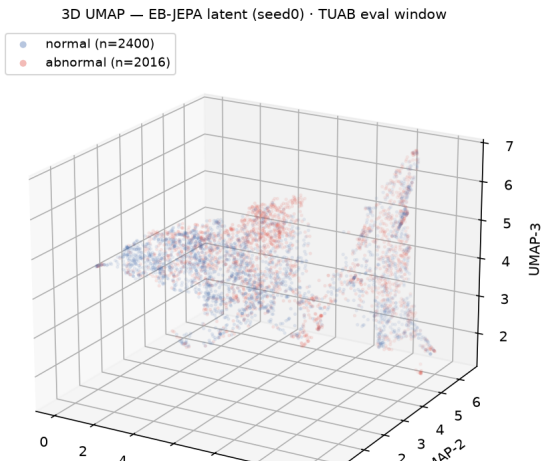
BAcc 0.802 (per-win) — sleep+pathology

pretraining, 1.2 M



Both LDA: *fit on TUAB train (2,717 recordings), projected on*

Latent space in 3D — recovering structure 2D collapses



Setup. Same encoder, window-level latents (seed 0), 3D UMAP instead of 2D. 2,400 normal / 2,016 abnormal windows.

★ **FINDING** A third dimension recovers structure
The 2-D UMAP panel (Latent Space slide, previous) looks mixed. In 3D the same embedding shows a two-lobe split that the 2-D projection collapses

CONCLUSION

What we learned — and what's next

What works

- ▶ **Corruption augmentation** (mask + spikes): +2.3 pp BAcc, 3-seed robust.
- ▶ **Weak invariance** $\lambda_{\text{inv}}=1$: +2–3 pp vs. VICReg's vision default ($\approx 6\sigma$), EEG-specific.
- ▶ **SIGReg** $\geq$ **VICReg**: best per-window + best AUROC.
- ▶ **TUEV transfer 0.425**: frozen, no TUEV labels, beats two supervised baselines.

What doesn't move the needle

- ▶ Scaling the encoder (0.4 M $\rightarrow$ 1.35 M): same plateau $\Rightarrow$ capacity is not the limit.
- ▶ 4 $\times$ more pretraining data (saturates).
- ▶ Spectral / stylised-facts terms beyond row 1's spec 0.1.
- ▶ EEGPT-style channel-pool transformer (collapsed under channel-drop).

Thank you

EB-JEPA for Clinical EEG — a first run

Team **vivatech-slightlyunawarefc** — questions welcome